

Comments from Reviewer 1:

Overall, this is a worthwhile study which I feel should be published. The authors essentially describe a substudy involving the diabetic patient (N=142) cohort from a previously published N=897 patient study (Besette et al, *BJC Reports*, 2025) which looked at the impact of comorbidities on patient survival. Here, they focus on type 2 diabetes (T2DM) and insulin use, including average daily dose of insulin used. The results appear to show that insulin-use does negatively impact on GBM patient survival, and should be considered as a negative prognostic indicator.

Response: We thank Reviewer 1 for his/her favorable opinion of our work.

Minor Comments:

- The title is not fully reflective of the content. "Insulin-use" rather than "Insulin" would be more accurate, and this change should be made throughout the manuscript.

Response: Thank you very much. We changed the title to "Insulin Use Negatively Impacts Glioblastoma Patient Survival."

- MGMT methylation % Cut off should be clarified.

Response: We thank Reviewer 1 for this comment and we respectfully disagree. The exact %cutoff used in the assay for MGMT methylation is most likely not relevant for the following reasons. First, the focus of our work is not on the accuracy of MGMT methylation status in predicting the survival of glioblastoma patients with and without diabetes. Instead, we are testing whether the methylation status is still prognostic in our diabetic population, since we have already shown that in our prior publication for the entire cohort (Besette et al, *BJC Reports*, 2025). Unfortunately, the smaller sample size in both cohorts (methylated n=39 [28%] and unmethylated n=31 [22%]) did not provide sufficient power for comparison. We think that this is impacted by the number of unknowns (n=71) which comprised more than half of the diabetic population (51%), compared to just 43% for the non-diabetic cohort. This is the reason for our previous sensitivity analysis, and it showed a trend suggesting that MGMT methylation status also influenced the diabetic population. However, this is not statistically robust as commented by another reviewer. We therefore incorporated a Cox proportional hazard analysis (Table 2) and still showed a trend for improved survival in the methylated subpopulation. Second, the cutoff value probably changed somewhat over time in our CLIA-certified molecular laboratory. To retrospectively adjust the MGMT status based on the most recent cutoff is probably not informative. Third, clinicians treat patients based on the MGMT status provided by the pathology department at different timepoints since this is also a predictive biomarker. Trying to adjust retrospectively while

discounting their management decisions may introduce additional errors and this practice may be counterproductive to our main objective. Lastly, we acknowledge that our analysis is retrospective and prospective validation is needed.

- “Table 2” is a combination of figures, not a table, and not very informative. Suggest revising.

Response: We revised Table 2 accordingly.

- Figure 2 B-F are too squashed and not reader-friendly. Suggest revised layout (content appears to be good).

Response: Thank you for pointing this out. We agree and revise Figure 2 accordingly.

- Figure 6 is not very informative or useful. Suggest change to supplementary Figure.

Response: We now combined Figure 6 and Supplementary Figure 4 in the first submission as Supplementary Figure 3 in the revision. The new supplementary figure has both tumor size and growth velocity information.

- Some minor spelling errors throughout – eg. “Acarbose”

Response: Thank you. We changed acarbose and a spell check.

- Results section – Some speculation and discussion points included which should be moved to the appropriate sections.

Response: We appreciate Reviewer 1’s comment on this. This may be a stylistic issue. We provided some speculation in the summary sentence of certain paragraphs in the Results section to help orient readers with the purpose of our analysis. We believe this will be easier for the readers to recall the data and the corresponding issues later in Discussion.

- Discussion should reflect that “Insulin-use” rather than “insulin” itself is what the study shows.

Response: We agree. “Insulin use” rather than “insulin” is provided in Discussion.

Major Comments:

- IDH mutation status. 10 IDH mutant patients included in non-diabetic population. These would no longer be consider WHO Grade 4 GBM, based on updated 2021 classification, and would be expected to have improved survival and therefore confound the results. These should be removed for accurate comparison. Similarly, the sensitivity analysis for IDH mutation status is IDHwt vs. IDH-unknown. Lines 204-211 & Supp Fig

1A add no value to the analysis or manuscript and are potentially misleading for readers.

Response: We agree. The IDH mutation analysis is removed.

- Multivariate analysis. The figures in Table 2 appear to show multiple univariate analyses rather than a multivariate analysis and this is not well described in the Methods or Results sections. Suggest revision for clarity. This is also the case throughout. In line 280, the authors suggest “Collectively, these data indicate insulin’s negative effect on glioblastoma patient survival is dose-dependent and it is inversely related to the dose administered” – I do not believe they have provided sufficient statistical evidence to back up this claim given the potential for confounding factors in this population (42% KPS <60, extent of tumour resection, age, etc.). Given the granularity of the data that is likely available to the authors (based on the 2025 comorbidities paper), a deeper dive into the potential confounders in this population is warranted.

Response: In the new Table 2, we provided Cox proportional analysis to address this criticism. In addition, we performed a permutation analysis to guard against type I error from multiple testing. We still find higher average daily dose of insulin is associated with a shortened survival.

- Insulin-Use section in Results. Paragraph needs to be re-worded for clarity and better described in the Methods section. Pre-planned or post hoc analysis? In the comparative analyses, are both Insulin & NIDM included/excluded? And if included, in which arm? The results section also includes background and discussion points.

Response: We thank Reviewer 1 for this question and an opportunity to clarify these points. First, this is not a prospective study and therefore the terms “pre-planned analysis” and “post hoc analysis” are not relevant. Second, we defined insulin-using diabetics as those using insulin with or without NIDM, while non-insulin-using diabetics as those using no insulin with or without NIDM.

Comments from Reviewer 2:

Summary

This retrospective single-institution study evaluates the impact of type 2 diabetes mellitus (T2DM) and insulin use on survival in 867 glioblastoma patients (142 diabetics; 725 non-diabetics) treated between 2005–2022. The authors report:

Shorter overall survival (OS) in diabetics vs non-diabetics (7.9 vs 12.3 months).

Insulin use—rather than diabetes per se—associates with worse survival.

A dose-dependent association between insulin exposure and OS.

No correlation between insulin dose and tumor size or growth velocity.

The topic is clinically relevant and biologically plausible (PI3K/insulin signaling axis), and the dataset is reasonably large overall. The manuscript is clearly written and the figures are extensive.

[Response: We thank Reviewer 2 for his/her favorable overall comment on our work.](#)

However, major methodological concerns limit causal interpretation, particularly regarding confounding by indication, time-dependent bias, missing data handling, and statistical modeling strategy. These issues must be addressed before publication.

[Response: We appreciate your thoughtful comments and will strive to address your concerns.](#)

Major Comments

1. Confounding by Indication (Critical)

The central claim is that insulin negatively impacts survival. However:

Insulin use likely reflects more severe diabetes, poorer glycemic control, higher steroid exposure, or systemic frailty.

[Response: We thank Reviewer 2 for this comment. We went back to our dataset and evaluated the number of diabetic complications and glycosylated hemoglobin levels between our insulin-using and non-insulin-using diabetics. We found that diabetics who used insulin had higher HbA1c levels. However, we did not find any statistical difference in the number of diabetic complications between the two groups, suggesting that insulin-using diabetics probably have more advanced insulin resistance compared to those not using insulin. The shortened survival in insulin-using diabetics is not from the number of diabetic complications. This is pointed out in both Results and Discussion sections.](#)

Diabetics were older and had lower KPS.

[Response: Diabetes is a result of aging. T2DM is related to aging, and the prevalence increases from 3.0% among those at age 18 to 44, to 14.5% in those 45 to 64, and 24.4% in those 65 or older \(reference 11\).](#)

[Therefore, it is expected that our patients with diabetes are older and have lower KPS than non-diabetics. Age and KPS were also covariates incorporated in our Cox](#)

proportional hazard analyses for both non-diabetics and diabetics (see Figures 3B, 4B and 5B), and significant hazard ratios were noted in either group after adjustment. Therefore, glioblastoma patients with diabetes have shorter survival compared to non-diabetics is not necessarily due to older age or poorer KPS. This is addressed in Discussion.

Insulin-treated patients may have more advanced comorbidity burden.

Response: To keep our focus on diabetes-related co-morbidities, we did not find any statistical difference, suggesting that the shortened survival in insulin-using diabetics is not from the number of diabetic complications (please see Supplementary Figure 2A).

The current analysis does not adequately adjust for:

Age (continuous)

KPS (continuous)

Extent of resection

MGMT status

Steroid exposure (critical in GBM)

Comorbidity burden

Treatment intensity

Although Cox models are presented, insulin use is not shown in a fully adjusted multivariable model incorporating all established GBM prognostic variables.

Required:

Multivariable Cox model including:

Age (continuous)

KPS (continuous)

Resection

MGMT

Treatment

Diabetes status

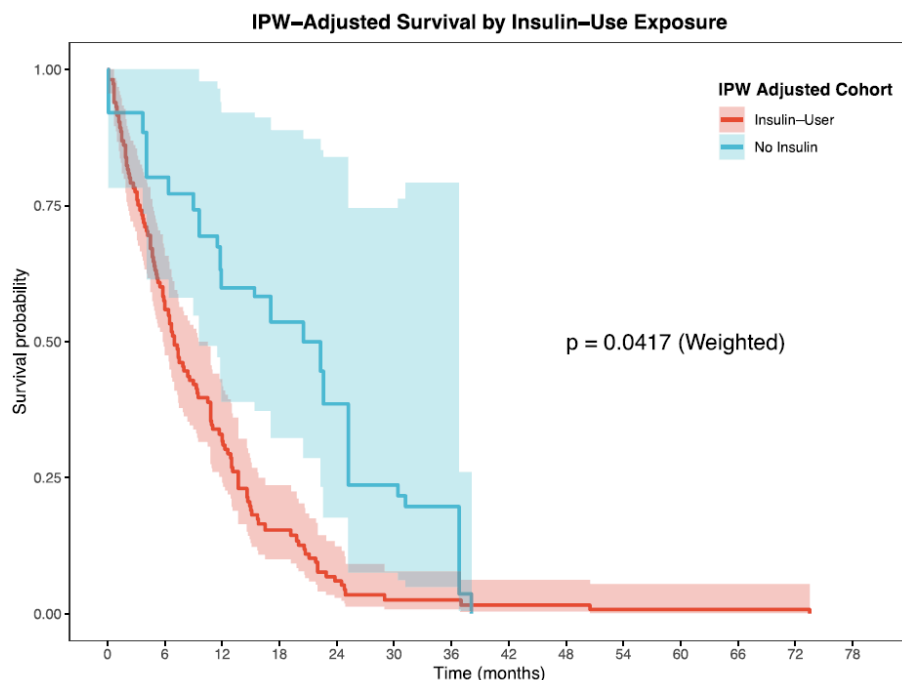
Insulin use

Ideally, propensity score matching or inverse probability weighting for insulin exposure.

Without this, the conclusion that “insulin negatively impacts survival” is not sufficiently supported.

Response: We thank Reviewer 2 for pointing this out. In Figures 3, 4 and 5, we performed two Cox proportional multivariate analyses adjusted for (i) age, KPS and resection and (ii) age KPS, resection, MGMT and treatment. We did both because (i) age KPS and extent of resection were significant prognostic factors in our cohort with diabetes, while (ii) age KPS, resection, MGMT and treatment were significant in our non-diabetic cohort. In analysis (i) we found that the hazard ratio disfavored diabetics in Figure 3, insulin-using diabetics in Figure 4, and diabetics who used insulin after glioblastoma diagnosis in Figure 5, all with 95%CI >1. In contrast, in analysis (ii) we found that the hazard ratio favored non-diabetics in Figure 3, non-insulin-using diabetics in Figure 4, and diabetics who did not use insulin after glioblastoma diagnosis in Figure 5, all with 95%CI <1. We conjecture that this is a result of a sample size not large enough to definitively separate out the two groups in multivariate analysis. But we are still able to identify significance in either group independently when adjusted using their respective prognostic factors.

We also performed an inverse probability weighted analysis, with adjustment for age, KPS, resection status, treatment, MGMT methylation status, sex, average steroid dose used, and non-insulin diabetes medication(s) use. We found that glioblastoma patients who used insulin still had worse survival compared to those who did not use insulin ($p=0.0417$) (see below and data not shown in publication). Therefore, we believe our conclusion that “insulin negatively impacts survival” is still valid in our analysis.



Regardless, our work here is still hypothesis generating that requires validation. Indeed, we pointed out in Discussion that we will need to confirm our findings using a validation dataset.

2. Immortal Time Bias / Time-Dependent Exposure

Insulin exposure appears treated as a fixed baseline variable. However:

Some patients may initiate insulin after diagnosis.

Insulin dose may change over time.

This introduces potential immortal time bias.

Required:

Clarify whether insulin use was defined at diagnosis.

If not, perform time-dependent Cox modeling.

Response: We thank Reviewer 2 for pointing this out. We performed a time-dependent Cox proportional hazard analysis based on whether patients used insulin before, after, or both before and after diagnosis of their glioblastoma (this is shown in Figure 5B). We found that diabetic patients used insulin before and after diagnosis (+/+) had a significantly increased hazard ratio, while those used after diagnosis (-/+) had a trend for increased hazard, after adjusting for age, KPS and extent of resection (upper panel in Figure 5B). In contrast, diabetes who did not use insulin behaved like non-diabetics

and their hazard ratio dropped significantly when adjusted for age, KPS, resection, MGMT and treatment (lower panel in Figure 5B). Therefore, we conclude that insulin use has a significant survival impact on glioblastoma patients with diabetes.

3. Insulin Dose Cutoff Optimization (Statistical Overfitting)

The manuscript identifies the optimal cutoff for insulin dose by iteratively testing multiple thresholds and selecting the lowest p-value.

This is effectively data-driven dichotomization, which inflates type I error.

Required:

Adjust for multiple testing (e.g., permutation test).

Or analyze insulin dose as a continuous variable in Cox regression.

Avoid dichotomization unless pre-specified.

The current approach likely overstates dose dependence.

Response: We thank Reviewer 2 for pointing this out. We respectfully disagree with him/her on our dichotomization methodology. Our work is a retrospective, and not a prospective, analysis. Therefore, we chose dichotomization as a means for exploring whether insulin dose makes a survival difference in glioblastoma patients with diabetes. Reviewer 2 is correct that this will inflate type 1 error, and our finding here will require confirmation in a larger dataset. It is worth noting that insulin dose on glioblastoma survival cannot be tested prospectively because (i) giving more insulin can potentially have a detrimental effect, and (ii) no patient will sign up for a clinical trial knowing that the objective is to test whether insulin will shorten his or her survival. Therefore, confidence in insulin dose-survival relationships can only be obtained based on concordance from multiple datasets.

Per Reviewer 2's comment, we also performed permutation testing on our dichotomization and, as expected, found that the p value both increased with respect to average daily insulin dose and cumulative insulin dose. The p value from permutation testing for the cutoff value of 10 units/day was still significant ($p=0.0170$). However, the p value for the cumulative dose cutoff was not ($p=0.2290$). This is in line with our prior analysis of survival from daily and cumulative dexamethasone doses where daily doses are more correlative with survival than cumulative doses (reference 24 Wong ET, et al. *Br J Cancer* 2015).

4. Missing Data (MGMT and IDH)

MGMT missing in 51% of diabetics is substantial.

The “sensitivity analysis” where unknowns are assigned to either group is not statistically robust.

Required:

Use multiple imputation for missing MGMT.

Provide missingness patterns.

Response: We agree with Reviewer 2 that missing MGMT data is substantial, and our prior sensitivity analysis was not statistically robust. Therefore, we took out this analysis and directly compared between methylated and unmethylated patients in Table 2.

Clarify whether MGMT testing varied by era (2005–2022).

Response: This is hard to trace back because MGMT testing was performed under the guidance of different laboratory directors. Regardless, MGMT was performed in a CLIA-certified laboratory and we believe that it was done using methods accepted during that period. Furthermore, we believe that neuro-oncologists or oncologists treating glioblastoma patients most likely will give temozolomide anyway for those with unmethylated MGMT during radiation due to a lack of alternatives, as long as the patient is fit to receive chemotherapy.

IDH unknown status (45%) also suggests temporal heterogeneity.

Response: As suggested also by Reviewer 1, we took out the IDH analysis because of our focus is on the IDH wild-type glioblastoma population.

5. Small Subgroup Sizes

Non-insulin diabetics (n=22)

Non-insulin medication only (n=10)

Neither medication (n=12)

Tumor growth velocity subset (n=35)

These small subgroups significantly limit inference.

Hazard ratios with wide CIs (e.g., Figure 4B) suggest instability.

The manuscript should temper conclusions accordingly.

Response: We agree with Reviewer 2 that the above subgroups have smaller sample sizes. This is a limitation of the dataset and we pointed this out in Discussion that our findings will require validation in a larger dataset.

6. Use of Pearson Correlation for Survival

Pearson correlations between age and survival are not ideal because survival is censored data.

Kaplan-Meier and Cox regression are appropriate; Pearson r is not optimal here.

This analysis should be removed or reframed.

Response: We appreciate Reviewer 2 for pointing out problems with censored survival data when performing a Pearson correlation between age and survival in Figure 1. However, the point here is to compare age and survival in non-diabetic and diabetic glioblastoma patients. We agree with Reviewer 2 that there are limitations, but we believe it is acceptable as long as the same methodology is used in both groups. Furthermore, Kaplan-Meier statistics were performed for the prognostic significance of age in the diabetic subgroup and showed significance that older age has a shortened median survival. Therefore, we respectfully disagree with Reviewer 2 and keep 1C and 1D in Figure 1.

7. Interpretation of Tumor Growth Analysis

The absence of correlation between insulin and tumor size does not negate a systemic effect on survival.

The discussion suggests mechanistic interpretations (inflammation, interstitial glucose) without supporting data.

This section should be toned down.

Response: We respectfully disagree with Reviewer 2 on this point. We did not offer a biological, mechanistic interpretation on the lack of correlation between insulin and tumor size. Rather, the method of measuring tumor size has limitations due to microscopic infiltration of glioblastoma cells into the surrounding brain. This is the reason why RANO 2.0 uses volumetrics as a measure of tumor size. Still, we believe that RANO 2.0 cannot account for microscopic tumors.

Minor Comments

KPS estimation from clinical notes may introduce misclassification bias.

Response: Unfortunately, this is a limitation of retrospective analysis.

Clarify whether HbA1c data were available.

Response: HbA1c data were available and we subsequently extracted and analyzed them. We found that diabetics who used insulin had higher HbA1c levels. However, we

did not find any statistical difference in the number of diabetic complications between the insulin-using and non-insulin-using groups, suggesting that insulin-using diabetics probably have more advanced insulin resistance compared to those not using insulin. The shortened survival in insulin-using diabetics is not from the number of diabetic complications.

Clarify steroid exposure (very important confounder).

Response: We thank Reviewer 2 for pointing this out. We did not find a survival difference between high versus low dexamethasone usage (Supplementary Figure 2C). However, our focus here is insulin use. Although there are other drugs potentially that can normalize glucose, insulin seems to be the quickest and most effective way of fixing hyperglycemia. This is pointed out in Discussion.

“Median OS 11.8 (95%CI 11.5–9.0)” appears reversed in one section.

Response: We thank Reviewer 2 for pointing this out. It has been fixed.

Some p-values are from Kruskal-Wallis applied to survival data; log-rank test should be used instead.

Response: Log-rank test was applied to survival data.

Figure density is high; consider consolidating supplementary figures.

Response: The figures have been modified accordingly.

Strengths

Large overall cohort (n=867).

Biologically plausible hypothesis.

Detailed medication extraction.

Dose-response exploration (conceptually strong).

Transparent R code repository.

Interpretation

The study convincingly shows:

Diabetics have worse survival than non-diabetics.

Insulin use correlates with worse outcomes.

Response: We thank Reviewer 2 for his/her favorable comments.

However, it does not yet convincingly demonstrate that insulin independently drives mortality rather than reflecting disease severity or metabolic frailty.

The current version supports association, not causation.

Response: We thank Reviewer 2 for his/her comment. Indeed, we clearly stated these limitations in the Discussion section and pointed out that there is an association between insulin use and poor patient survival. We also minimized various confounders as best as we could throughout our analysis.

Comments from Reviewer 3:

This is a re-analysis of a prior published database of GBM patients. The authors present findings of lower survival in diabetic patients and, amongst diabetics, lower survival for those on insulin vs not. To my mind it is difficult to account for confounding factors in a retrospective study such as this - patients with diabetes will be less well and therefore likely to have worse survival in any event and those on insulin are even less well at baseline. Indeed on multivariate analysis in the original the original published dataset (Figure 5, Bessette et al, Oct 2025), diabetes was not an independent prognostic factor. Could consider re-analysis of the diabetic population alone with a multivariate analysis in an effort to account for confounding factors to try show some evidence that insulin is an independent detrimental factor here but even then it's hard to fully account for confounding bias.

Response: We thank Reviewer 3 for pointing this out. This is precisely the point of our work: glioblastoma patients with diabetes do not have a shortened survival because of diabetes *per se*. It is a result of their insulin use. Our work, albeit retrospective in nature, provides compelling evidence that insulin is the culprit, but this finding will require confirmation using a validation dataset.

We also would like to acknowledge that we cannot test prospectively the insulin dose on glioblastoma. This is because (i) giving more insulin can potentially have a detrimental effect, and (ii) no patient will sign up for a clinical trial knowing that the objective is to test whether insulin will shorten his or her survival. Therefore, confidence in insulin dose-survival relationships can only be obtained based on concordance from multiple datasets.